MAT3280 Midterm exam – sample solution

11/02/2024

Question 1. (15 points) Let $\mathscr{F}$ be the σ-field on $\mathbb{R}$ generated by closed intervals in the form $[n, n+1]$, where $n \in \mathbb{Z}$ is an integer. Determine whether the following sets belong to $\mathscr{F}$.

(a) closed interval $[a, b]$, where $a < b$ are both integers.

(b) a set consisting of a single point, $\{c\}$, for an integer c.

(c) open interval (a, b), where $a < b$ are both integers.

Solution. (a) We can construct $[a, b]$ by taking union

$$[a, b] = [a, a+1] \cup [a+1, a+2] \cup \cdots \cup [b-1, b].$$

All closed intervals on the right are generators. The union of them must be in $\mathscr{F}$.

(b) Write $\{c\}$ as an intersection

$$\{c\} = [c-1, c] \cap [c, c+1].$$

(c) From (a) we know that the closed interval $[a, b]$ is in $\mathscr{F}$. We can take away the two boundary points by

$$(a, b) = [a, b] \cap \{a\}^c \cap \{b\}^c.$$

The two sets $\{a\}$ and $\{b\}$ are in $\mathscr{F}$, and so are their complements.

Question 2. (15 points) Let n denote a positive integer, and Ω be the set $\{1, 2, 3, \ldots, 2n\}$. Consider the collection of sets $\mathscr{D}$ that consists of subsets of Ω with even cardinality. This means that a set in $\mathscr{D}$ has size $2s$, where $s = 0, 1, 2, \ldots, n$.

(a) Show that $\mathscr{D}$ is a λ-system.

(b) For what value of n is $\mathscr{D}$ an algebra?

Solution. (a) We check the three defining properties of λ-system.

(i) The sample space Ω is in $\mathscr{D}$, because $|\Omega|$ is an even number.

(ii) If A has even cardinality, say $2s$, then the complement of A in Ω has cardinality $2n - 2s$, which is an even number.

(iii) Because Ω is finite, we only need to show that $\mathscr{D}$ is closed under disjoint union. Suppose A and B are disjoint sets in $\mathscr{D}$. Let $|A| = 2r$ and $|B| = 2s$. Since A and B are disjoint, the union $A \cup B$ has size $2r + 2s$, which is an even number. Therefore $A \cup B$ is in $\mathscr{D}$.

(b) For $n > 1$, the system $\mathscr{D}$ is not closed under intersection. For example, we can ahve

$$\{1, 2\} \cap \{2, 3\} = \{2\}.$$

$\mathscr{D}$ is a σ-algebra when and only when $n = 1$.

Question 3. (15 points) Let X be a continuous random variable with exponential distribution and mean 1. The probability density function of X is

$$f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & x \leq 0. \end{cases}$$

(a) Let Y denote the random variable $\min(X, 3)$. Derive the cumulative distribution function (cdf) of Y, i.e., find $P(Y \leq y)$ for $y \geq 0$.

(b) Let $\alpha(x)$ denote the answer in part (a), and let μ be the Lebesgue-Stieltjes integral induced by $\alpha(x)$. Evaluate the measure of the singleton $\{3\}$ with respect to measure μ.

(c) Compute the Riemann-Stieltjes integral $\int_0^\infty x\, d\alpha(x)$.

Solution. (a) For $0 \leq y < 3$,

$$P(Y \leq y) = \int_0^y e^{-x}\, dx = 1 - e^{-y}.$$

When $y \geq 3$, $P(Y \leq y)$ is a constant

$$P(Y \leq y) = 1.$$

In summary, the cdf of Y is

$$P(Y \leq y) = \begin{cases} 1 - e^{-y} & \text{if } 0 \leq y < 3 \\ 1 & \text{if } y \geq 3. \end{cases}$$

(b) The probability of the event $\{3\}$ equals to how much $\alpha(x)$ jump at the point 3, and it is

$$1 - (1 - e^{-3}) = e^{-3}.$$

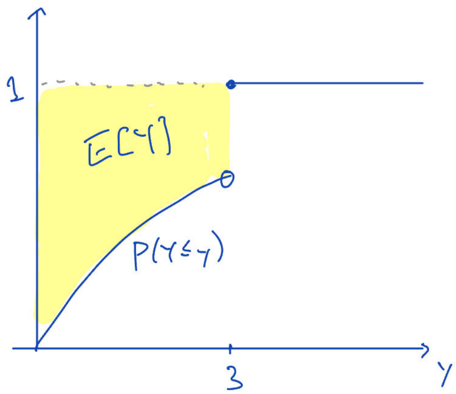
(c)

$$\begin{aligned} \int x\, d\alpha(x) &= \int_0^3 xe^{-x}\, dx + 3e^{-3} \\ &= [-xe^{-x}]_0^3 + \int_0^3 e^{-x}\, dx + 3e^{-3} \\ &= \int_0^3 e^{-x}\, dx \\ &= 1 - e^{-3}. \end{aligned}$$

Alternately, we can apply the formula

$$E[Y] = \int_0^\infty (1 - F_Y(y))\, dy.$$

when X is a nonnegative random variable. The expectation is the area bounded between the cdf and the horizontal line with y-intercept 1.



The shaded region has area

$$\int_0^3 e^{-x}\,dx = 1 - e^{-3}.$$

Question 4. (15 points) Let $A_1, A_2, A_3, \ldots$, be a sequence of events in a probability space $(\Omega, \mathscr{F}, P)$. Suppose $\lim_{k\to\infty} P(A_k)$ exists and is equal to L.

Consider the event $B = \{\omega \in \Omega : \omega$ occurs in A_k for all but finitely many $k\}$, i.e.,

$$B = \{\omega \in \Omega : \exists N, s.t.\ \omega \in A_k \text{ for all } k \geq N\}.$$

(a) Show that $P(B) \leq L$

(b) Assume further that $A_1 \supseteq A_2 \supseteq \cdots$ is a decreasing sequence of sets. Show that $P(B) = L$.

Solution. (a) The event B is the liminf of A_k,

$$B = \liminf_k A_k.$$

The indicator function of B is the liminf of the indicator functions of the sets A_k.

By Fatou lemma,

$$\int \liminf_k \mathbf{1}_{A_k}\,dP \leq \liminf_k \int \mathbf{1}_{A_k}\,dP.$$

The left-hand side is the probability $P(B)$, and the right-hand side is equal to L, because it is assumed that the limit of $P(A_k)$ exists, and the limit is equal to the liminf.

(b) When the sequence of sets A_k is decreasing, the liminf is equal to the intersection

$$B = \liminf_k A_k = \cap_k A_k.$$

We obtain $P(B) = L$ by appealing to the upper semi-continuity of probability measure. Indeed, we have

$$L = \lim_{k\to\infty} P(A_k) = P(\cap_k A_k) = P(B).$$

Question 5. (20 points) In this question, we consider a random source that produces an infinite sequence of uniformly random bits. Denote the bits in the sequence by B_k, for $k = 1, 2, 3, \ldots$. Assume that they are independent and $P(B_k = 0) = P(B_k = 1) = 1/2$, for $k = 1, 2, 3, \ldots$.

Given the infinite random sequence of bits, we produce another sequence of nonnegative integers as follows: let X_1 denote the position of the first "1", and for $i \geq 2$, let X_i be defined as the difference between the position of the $(i+1)$-th "1" and the position of the i-th "1". For instance, suppose the random sequence starts with $0001101000\ldots$. Then $X_1 = 4$, since the $k = 4$ is the first index such that $B_k = 1$. The value of X_2 is 1, because the next 1 occurs at position 5, and $5 - 4 = 1$. Since the third 1 occurs at position 7, the next value in the sequence is $X_3 = 7 - 5 = 2$.

(The definition can be put more concisely by defining $B_0 = 1$. Then X_i is the distance between the i-th 1 and the $(i + 1)$-st 1, for $i \geq 1$.)

(a) Find the distribution of X_i. That is, for any integer $j \geq 0$, evaluate the probability $P(X_i = j)$.

(b) What is the probability that there are 100 consecutive numbers in the sequence $(X_i)_{i=1}^{\infty}$ that are strictly increasing?

(c) What is the probability that the sequence $(X_i)_{i=1}^{\infty}$ does not contain a particular positive integer? That is, find the probability of the event "$\exists n \in \mathbb{N}$ *s.t.* $\forall i$ $X_i \neq n$".

Solution. (a) The sample space Ω for this question is the set of all infinite binary sequences. The σ-field is generated by events in the form

$$\{B_1 = b_1, B_2 = b_2, \ldots, B_m = b_m\}$$

for $m \geq 1$, and for $b_1, \ldots, b_m \in \{0, 1\}$. The probability measure P is defined such that $(B_k)_{i=1}^{\infty}$ are independent and $P(B_k = 0)$ and $P(B_k = 1)$ are both equal to 1.

For every $i \geq 1$, X_i is a function mapping an infinite binary sequence in Ω to the set of natural numbers. The random variable X_i is geometrically distributed. For $j \geq 1$, X_i is equal to j if we see the pattern

$$\underbrace{0000}_{j-1}1$$

after the i-th 1. Since all the bits are equally likely,

$$P(X_i = j) = \frac{1}{2^j}$$

for $j \geq 1$.

(b) We apply BC2 to see that the answer is 1. We will only consider the special increasing numbers $1, 2, 3, \ldots, 100$. If this pattern occurs with probability 1, then it is already sufficient to show that the probability in the question is also 1.

Divide the time indices for the sequence $(X_i)_{i=1}^{\infty}$ into disjoint groups of size 100, from 1 to 100, 101 to 200, 201 to 300, etc. The probability that the numbers 1 to 100 appear consecutively is

$$p = \frac{1}{2} \cdot \frac{1}{2^2} \cdot \frac{1}{2^3} \cdot \frac{1}{2^4} \cdots \frac{1}{2^{100}}.$$

Since p is a positive constant, and what happens in a group is independent are mutually independent, we can apply Borel-Cantelli lemma 2 to conclude that, with probability 1, we can see the numbers 1 to 100 infinitely often in a group. So in particular, this event occurs at least once with probability 1.

(c) Let n be an integer. The events $\{X_i = n\}$, for $i \geq 1$, are independent and each of them has probability $1/2^n$. Because the sum of probability $P(X_i = n)$ is infinity, by Borel-Cantelli lemma 2, $X_i = n$ infinitely often with probability 1.

In this question we only need to consider the event $X_i = n$ for at least one i. For $i \geq 1$, let A_n denote the event that $X_i = n$ for at least one index i. In the previous paragraph we see that $P(A_n = 1)$ for all $n \in \mathbb{N}$. Since the intersection of a countable intersection of sure event is also a sure event,

$$P((X_i)_{i=1}^{\infty} \text{ contains all positive integers}) = P(\cap_n A_n) = 1.$$

Therefore, the probability that the sequence $(X_i)_i$ does not contain a particular positive integer is 0.

Question 6. (20 points) Let $(\Omega, \mathscr{F}, \mu)$ be a measure space, and $X(\omega)$ be a nonnegative measurable function. (The integral $\int X\,d\mu$ may or may not be finite.) Define a set function

$$\nu(E) \triangleq \int \mathbf{1}_E X\,d\mu$$

for $E \in \mathscr{F}$. (The symbol $\mathbf{1}_E$ represents the indicator function of E.)

(a) Prove that $(\Omega, \mathscr{F}, \nu)$ is a measure space.

(b) Show that for any simple function $f(\omega)$, $\int f\,d\nu = \int fX\,d\mu$.

(c) Show that for any nonnegative measurable function $g(\omega)$, $\int g\,d\nu = \int gX\,d\mu$.

(d) Show that for any ν-integrable function $h(\omega)$, $\int h\,d\nu = \int hX\,d\mu$.

Solution. (a) We only need to show that ν is a measure. We check the axioms of measure function.

(i) Suppose E is the empty set. The function $\mathbf{1}_\emptyset X$ is the zero function. Hence,

$$\nu(\emptyset) = \int 0\,d\mu = 0.$$

(ii) Let E_1, E_2, $E_3, \dots$ be a sequence of mutually disjoint $\mathscr{F}$-measurable sets. For any finite integer n, we have

$$\begin{aligned}
\nu(E_1 \cup \cdots \cup E_n) &= \int \mathbf{1}_{E_1 \cup \cdots \cup E_n} X\,d\mu \\
&= \int (\mathbf{1}_{E_1} + \cdots + \mathbf{1}_{E_n}) X\,d\mu \\
&= \sum_{k=1}^{n} \int \mathbf{1}_{E_k} X\,d\mu \\
&= \sum_{k=1}^{n} \nu(E_k).
\end{aligned}$$

For countable union, we apply monotone convergence theorem, and consider the sequence of functions $(f_n)_{n=1}^{\infty}$, with f_n defined by

$$f_n \triangleq \mathbf{1}_{E_1} + \cdots + \mathbf{1}_{E_n}.$$

Because the indicator function are nonnegative, $(f_n)_{n=1}^{\infty}$ is an increasing sequence of functions. Moreover, if we set $E = \cup_{n=1}^{\infty} E_n$, the indicator function $\mathbf{1}_E$ is equal to $\sum_{n=1}^{\infty} \mathbf{1}_{E_n}$.

We thus have

$$\begin{aligned}
\nu\big(\cup_{n=1}^{\infty} E_n\big) = \nu(E) &= \int \mathbf{1}_E X \, d\mu \\
&= \int \sum_{n=1}^{\infty} \mathbf{1}_{E_k} X \, d\mu \\
&= \sum_{n=1}^{\infty} \int \mathbf{1}_{E_k} X \, d\mu \\
&= \sum_{n=1}^{\infty} \nu(E_n).
\end{aligned}$$

We note that the second last equality is due to MCT. This proves that ν is σ-additive.

(b) A simple function $f(\omega)$ can be expressed as a finite sum

$$f(\omega) = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}(\omega),$$

for some real number a_k and $\mathscr{F}$-measurable set A_k, for $k = 1, 2, \dots, m$.

Using this representation, we obtain

$$\begin{aligned}
\int f \, d\nu &\triangleq \sum_{k=1}^{m} a_i \nu(A_k) \\
&= \sum_{k=1}^{m} a_i \int \mathbf{1}_{A_k} X \, d\mu \\
&= \int (\sum_{k=1}^{m} a_i \mathbf{1}_{A_k}) X \, d\mu \\
&= \int f X \, d\mu.
\end{aligned}$$

The second last step is the linearity property for simple function.

(c) We first find a sequence of simple function $g^{(n)}$ that converges to g from below. We know that this can always be done (see the construction in (6.4) in textbook). The argument is a reduction to simple functions using MCT.

$$\begin{aligned}
\int g\,d\nu &= \int \lim_n g^{(n)}\,d\nu \\
&= \lim_n \int g^{(n)}\,d\nu \\
&= \lim_n \int g^{(n)} X\,d\mu \\
&= \int \lim_n g^{(n)} X\,d\mu \\
&= \int gX\,d\mu.
\end{aligned}$$

The exchange of the order of limit and integral can be justified by MCT, because X is nonnegative, and $g^{(n)}X \nearrow X$

(d) When h is ν-integrable, we have

$$\int h\,d\nu \triangleq \int h^+\,d\nu - \int h^-\,d\nu.$$

By part (c),

$$\int h\,d\nu = \int h^+ X\,d\mu - \int h^- X\,d\mu.$$

We note that h^+X and h^-X are the positive part and negative pat of hX, respectively. Again, this requires the assumption that X is a nonnegative function. Finally, we obtain

$$\int h\,d\nu = \int hX\,d\mu.$$